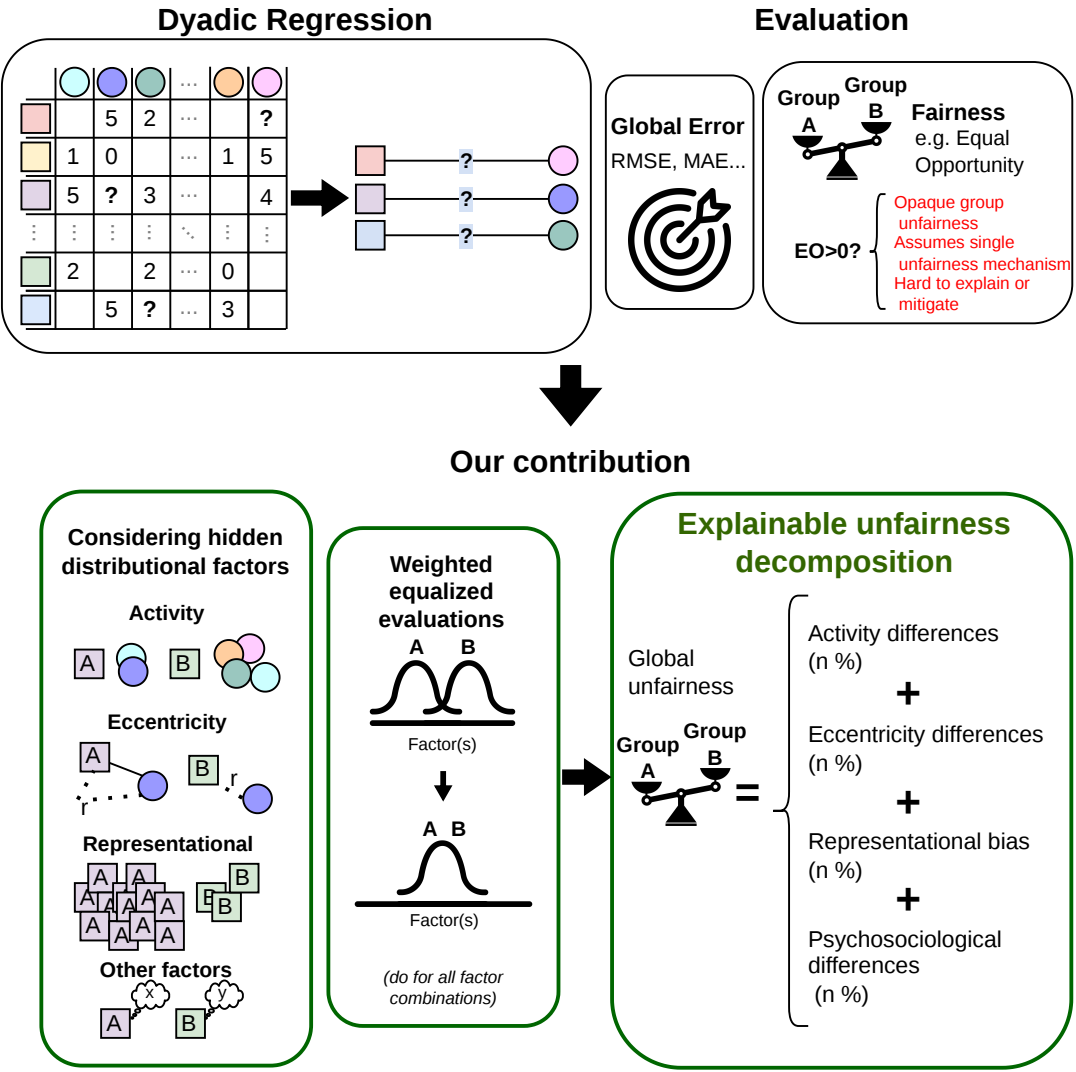

Dyadic Regression
Evaluation
Global Error
RMSE, MAE...
Group
A
Group
B
Fairness
e.g. Equal Opportunity
EO>0?
Opaque group unfairness
Assumes single unfairness mechanism
Hard to explain or mitigate
Our contribution
Considering hidden distributional factors
Activity
Eccentricity
Representational
Other factors
Weighted equalized evaluations
A
B
Factor(s)
A B
Factor(s)
(do for all factor combinations)
Explainable unfairness decomposition
Global unfairness
Group
A
Group
B
Activity differences (n %)
Eccentricity differences (n %)
Representational bias (n %)
Psychosociological differences (n %)